1. Human Workflow Specification (Gold Standard): Richmond et al. (2020) Realist Review - Human Reviewer Architecture and Algorithm

1.1 Objective (what the human workflow is trying to achieve)

Richmond et al. (2020) conduct a realist review to build an explanatory programme theory of how educational interventions develop analytical and non-analytical clinical reasoning in undergraduate health professions education, explicitly addressing why interventions work, for whom, when, and under what circumstances.

- **Evidence (Abstract)**: "This realist review examines the way educational interventions develop analytical and non-analytical reasoning ability in undergraduate education… seeking not only to identify if an intervention works, but also understand the reasons why, for whom, and in what circumstances."
- **Evidence (Abstract)**: "The aim of this study is to develop understanding… when they do, for whom and in what circumstances."
- **Evidence (Methods)**: "Realist research data are analysed and interpreted to form context, mechanism, outcome configurations (CMOCs) and collectively form a programme theory."

This specification "unpacks" the human reviewer workflow that produced the paper's final conclusions, i.e., the sequence of reasoning operations, decision rules, and intermediate products.

2. Human "System Architecture" (who does what, with what tools, producing what artefacts)

2.1 Human roles

Richmond's workflow is not a single-person extraction; it is a team process with defined responsibilities and quality control.

- **Lead reviewer (Anna Richmond; AR)**: conducts background/scoping searches, runs database searches, performs initial CMOC coding, synthesises results, and drafts the manuscript.
- **Evidence (Methods)**: "Initial background literature searches… was undertaken by the lead author (AR)."
- **Evidence (Methods)**: "Initial CMOC coding was undertaken by AR…"
- **Evidence (Author Contributions)**: "AR developed the protocol, performed the searches, performed initial data extraction, synthesised the results…"

- **Research team (AR, NC, SG, WA, RP)**: provides domain expertise and consensus building for the IPT, checks coding consistency, and provides feedback on the developing theory and manuscript.
- **Evidence (Methods)**: "All members of the research team… were clinical teachers…"
- **Evidence (Methods)**: "AR developed consensus amongst the research team…"
- **Evidence (Methods)**: "…all 28 articles were checked for consistency by another reviewer (RP, SG or NC)."

2.2 Tooling and data management

Richmond's reasoning is operationalised via concrete tools for storing texts, coding contexts, and elaborating causal links.

- **NVivo**: used to store full texts and code contexts.
- **Evidence (Methods 2.2)**: "NVivo… used to store full texts and code contexts."
- **Microsoft Excel**: used to elaborate "which contexts affected the mechanisms and outcomes," i.e., to organise and refine cross-study causal structure.
- **Evidence (Methods 2.2)**: "Microsoft® Excel… was used to further elaborate which contexts affected the mechanisms and outcomes."
- **Governance standards**: PROSPERO registration and RAMESES standards are explicitly referenced as process/quality anchors for realist synthesis reporting.
- **Evidence (Methods)**: "The review was registered on PROSPERO…"
- **Evidence (Methods)**: "The RAMESES… standards… were referred to throughout the review."

2.3 Core artefacts

The workflow produces and iteratively refines the following artefacts:

- **Aim + Research Questions** (the controlling specification of what the review must explain)
- **IPT (Initial Programme Theory)** (a theory seed that defines what counts as relevant evidence)
- **Search specification** (databases, year constraints, and theory-derived search concepts/terms)
- **Screening/appraisal decisions** (inclusion rules tied to theory contribution and methodological rigour)
- **CMOC set** (coded configurations across included papers)
- **Cross-study pattern map** (recurrent CMOC patterns and their consolidation into key contexts/mechanisms)

- **Final programme theory** (results + figures + conclusions expressing "what works, for whom, when, why")

3. Human Algorithm (step-by-step operations, decision rules, and iterative loops)

Stage 0 — Problem framing: define realist explanatory intent and formal research questions

Input

- The educational and safety importance of developing clinical reasoning.
- The observed problem that interventions may not work uniformly across learners and contexts.

Operation (human reasoning)

- Establish a realist explanatory aim (mechanism- and context-sensitive causation).
- Derive explicit research questions from the aim.

Output

- A stated aim plus two research questions guiding the entire process.

Evidence (Introduction)

- Aim is explicitly articulated, followed by:

"The research questions that emerged from this aim were: (i) What educational interventions are effective…?, and (ii) When and why are they effective, for whom and in what circumstances?"

Stage 1 — IPT construction (theory seeding): create the initial explanatory model before large-scale screening

This stage is the primary "human import" step: humans define the conceptual target that later governs search and inclusion.

Input

- Scoping/background literature
- Expert opinion and researcher experience
- Team expertise as clinical teachers

- Complementary learning theories to structure how “context” will be sought

Operation (algorithmic sequence)

1. Background/scoping searches are conducted (by AR).
2. The team builds consensus on what kinds of educational interventions promote analytical vs non-analytical reasoning.
3. Outputs from steps (1)–(2) are used to draft initial CMOC hypotheses.
4. After research-team feedback, these drafts are consolidated into an Initial Programme Theory (IPT).
5. Additional learning theories are integrated to define “context levels” to be investigated (student, teacher, learning encounter/activity, organisational).

Output

- IPT + initial CMOC drafts + a context-level lens that directs later searching and synthesis.

Evidence (Abstract/Methods)

- “Literature from a scoping search, combined with expert opinion and researcher experience was synthesised to generate an initial programme theory (IPT).”
- “AR developed consensus amongst the research team…”
- “The outputs… led to the development of initial drafts of CMOCs… Following feedback… an IPT was constructed…”
- “Other relevant theories were also integrated into the IPT… Cognitive flexibility theory… Situativity theory…”

Stage 2 — Search strategy and execution: theory-driven retrieval, with deliberate term expansion

Input

- IPT themes/constructs (the theory seed determines what to retrieve)

Operation (algorithmic sequence + loop)

1. Conduct structured searches in MEDLINE, PsycINFO, ERIC, and CINAHL (May 2017), incorporating IPT themes.

2. Apply a time boundary: searching begins in 2000, justified by the 1999 “To err is human” report.
3. Perform a supplementary search to add additional terms as initial searching identifies relevant constructs (pattern recognition, deliberate practice, illness scripts, knowledge acquisition/recall).
4. Operationalise “knowledge” as a multidimensional construct (Table 1) to support consistent reasoning about “low/high knowledge” contexts.

Output

- A candidate corpus for screening and appraisal.

Evidence (Methods 2.1)

- Database list, search date, year constraint, and supplementary terms are explicitly stated.

Stage 3 — Screening and appraisal: select studies for theory contribution and assess methodological credibility

This stage formalises the key realist inclusion principle: studies are included because they contribute to theory building, not merely because they match a topic.

Input

- Candidate corpus from Stage 2

Operation

1. Title/abstract screening assesses relevance; studies are retrieved if they are deemed to contribute to theory building.
2. Full-text review assesses methodological rigour as credibility/trustworthiness.
3. Apply the review’s operational definition of “educational intervention” for full-text decisions.
4. Include studies describing a technique that can be integrated into an intervention (not only fully packaged interventions).
5. Search reference lists of included papers for additional theory-contributing studies.

Output

- The final included set that will drive programme theory refinement.

Evidence (Methods/Results)

- "studies retrieved if they were deemed to contribute to theory building."
- "Methodological rigour… credible and trustworthy… assessed at full text review."
- Quantitative flow (non-ambiguous):

"In total, 149 full texts were retrieved. Of these, 25 articles met the inclusion criteria, and two more were added from reference list searching… One was added…"

⇒ 28 papers contributed.

Not reported in paper

- A named appraisal instrument (e.g., CASP) or a numerical scoring rubric is not specified; rigour is stated as credibility/trustworthiness.

Stage 4 — Data extraction and CMOC coding: convert each included paper into structured causal explanations

Input

- 28 included full texts
- IPT and CMOC framing
- Data extraction forms

Operation (algorithmic sequence + inference rules)

1. For each included full text, human reviewers devise CMOCs.
2. Apply realist inference rules for latent constructs:
- Mechanisms (especially cognitive/emotional learner responses) may be implicit and therefore are theorised or inferred from learning theory and study data.
- Outcomes may also be theorised when not explicit.
1. Quality assurance:
- AR performs initial CMOC coding.
- A second reviewer checks all 28 articles for consistency.
1. Manage and structure coded information:
- NVivo is used for storing texts and coding contexts.
- Excel is used to further elaborate which contexts affect mechanisms and outcomes (i.e., to refine causal mapping).

Output

- A CMOC set across all included studies, with coded contexts and structured links between contexts, mechanisms, and outcomes.

Evidence

- "The CMOCs were devised for all included full texts."
- "Initial CMOC coding was undertaken by AR and all 28 articles were checked for consistency by another reviewer…"
- "Mechanisms… are not always explicit… can be theorised… or inferred…"
- "All possible outcomes may also not be explicit… outcomes can be… theorised…"
- NVivo/Excel statements in Methods 2.2.

Important representation detail

Richmond's Results present CMOC statements using a C / Mresource / Mresponse / O structure (e.g., in Context 1), reflecting their operational decomposition of mechanisms into resource and response in reporting.

Stage 5 — Cross-study synthesis and iterative refinement: derive recurrent CMOC patterns and consolidate into programme theory

This stage is where the final conclusions are produced: human reviewers aggregate and reconcile CMOCs across studies, iteratively revisiting earlier papers as theory evolves.

Input

- CMOC set across studies + IPT

Operation (pattern-finding + iterative loop)

1. Compare studies and identify recurrent patterns of CMOCs.
2. Recognise that different studies contribute differently (some highlight contexts, others mechanisms).
3. Iteratively re-analyse earlier studies in light of theories arising from later studies.
4. Maintain a theory-building stance (rather than hypothesis testing) due to under-reporting or distance of outcomes.
5. Produce consolidated outputs: key contexts and key mechanisms determining effectiveness.

Output

- A refined programme theory and its core explanatory components.
- In Results, the programme theory is operationalised via five key student-level contexts, each presented with CMOC statements and supporting evidence; teacher/organisational contexts are noted as rarely discussed.

Evidence (Methods/Results)

- "recurrent patterns of CMOCs were identified."
- "Studies identified earlier were re-analysed…"
- "Key contexts and mechanisms… produced as outputs…"
- "Five key contexts were identified…"

4. Public, verifiable outputs (what Richmond reports that can be used as gold-standard targets)

These are the observable outputs that a verification effort can directly target:

- Included set size and flow: 149 full texts retrieved → 28 included (25 + 2 + 1).
- Programme theory expressed via CMOCs: CMOCs collectively form the programme theory.
- Five key contexts (student level) explicitly enumerated in Results:
- low knowledge / low domain knowledge / inability to apply knowledge
- high domain-specific knowledge
- positive coping strategies / appropriate self-confidence/self-efficacy
- negative coping strategies / low self-confidence/self-efficacy
- different levels of knowledge within a group
- High-level conclusions about what is required to develop reasoning (knowledge acquisition/recall + clinical experience + practice in real/simulated environments) and how contexts (pre-existing knowledge, self-confidence) condition effectiveness.

5. What this specification clarifies

This document makes explicit how Richmond's team produced the paper's conclusions:

- They did not start with "28 papers" arbitrarily; they built an IPT first, then used it to guide search and inclusion.
- Inclusion decisions were theory-contribution driven and filtered by rigour.
- They transformed papers into CMOCs using a realist inference rule that allows theorising/inference for mechanisms/outcomes when implicit.

- They produced conclusions via cross-study pattern identification and an iterative re-analysis loop, culminating in a programme theory expressed through five key contexts and CMOC statements.